Chandigarh University

UNIVERSITY INSTITUTE OF COMPUTING

"Transforming Healthcare Operations through Interactive Dashboards and Data-Driven Insights"

MINOR-PROJECT REPORT

ON

Analytical Dashboards in Healthcare

Program Name: BSC (CS)

DATA INTERPRETATION

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Chandigarh University

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to **Chandigarh University** for including the minor project as a vital part of our academic curriculum. As part of this initiative, we were given the option to choose project topic on our own. I opted for the **Dashboard Making**, which turned out to be one of the most enriching academic decisions of my journey so far.

I am immensely grateful to **Mr**, **Rajat Kumar** for giving me the opportunity to work under her expert guidance. His in-depth knowledge of data analytics and healthcare operations, along with her constant support and mentorship, played a crucial role in shaping my understanding of the subject. Her encouragement and constructive feedback helped me navigate every phase of this project with clarity and confidence.

This project allowed me to explore the powerful role that **data analytics and visualization play in enhancing hospital operations**, optimizing patient care, and making data-driven decisions. It gave me a chance to work on real-time datasets, understand operational flows, and develop meaningful dashboards that can support medical decision-making. The exposure I gained during this period not only boosted my technical skills but also deepened my appreciation for how information systems contribute to the healthcare ecosystem.

I would also like to extend my thanks to the faculty and team for cooperation and support throughout my project. Their openness to share insights and data played a big role in the success of my learning.

Lastly, I am thankful to my family, friends, and mentors who motivated me throughout this journey. Their belief in me and constant encouragement kept me focused and driven.

Mr. Rajat Kumar

(Data Interpretation)

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1. Executive Summary

This report presents the comprehensive findings and experiences from a project . The project focused on developing data visualization skills using Tableau to interpret and present hospital data effectively.

During this period, I successfully created multiple healthcare dashboards, with primary focus on two comprehensive visualization projects: Admissions Statistics Dashboard and Disease Statistics Dashboard. These dashboards were designed to provide actionable insights for hospital management, enabling data-driven decision-making in-patient care and resource allocation.

The project demonstrates the critical role of data analytics in modern healthcare systems, showcasing how proper visualization can transform raw hospital data into meaningful insights that directly impact patient outcomes and operational efficiency. The work completed during this project highlights the growing importance of data analytics professionals in healthcare settings and the immense potential for technology-driven improvements in medical care delivery.

Key achievements include the development of interactive dashboards that visualize patient admission trends, disease prevalence patterns, and operational metrics, along with recommendations for integrating advanced clinical data analytics to further enhance healthcare delivery at the institution.

2. Introduction

The healthcare industry generates vast amounts of data daily, from patient records and treatment outcomes to operational metrics and resource utilization patterns. However, the true value of this data lies not in its volume but in the insights that can be extracted and applied to improve patient care, optimize operations, and enhance overall healthcare delivery.

Healthcare data analytics has emerged as a critical discipline that bridges the gap between raw medical data and actionable intelligence. By leveraging advanced analytical tools and visualization techniques, healthcare professionals can identify trends, predict outcomes, and make informed decisions that directly impact patient welfare and institutional efficiency.

This project project was undertaken to gain practical experience in healthcare data analytics, specifically focusing on the application of business intelligence tools like Tableau to create meaningful visualizations from hospital data. The project aimed to demonstrate how data analytics can be seamlessly integrated into healthcare operations to support evidence-based decision making.

The choice of Satguru Partap Singh Hospital, Ludhiana, as the project data provided an excellent opportunity to work with real-world healthcare data in a dynamic hospital environment. This experience offered insights into the practical challenges of healthcare data management and the transformative potential of effective data visualization in clinical settings.

The significance of this project extends beyond academic learning, as it addresses the growing demand for skilled data analysts in healthcare settings and demonstrates the practical applications of computer science principles in improving healthcare outcomes.

3. Literature Review

3.1 Evolution of Healthcare Analytics

Healthcare analytics has evolved significantly over the past decade, transforming from basic reporting systems to sophisticated predictive modelling and real-time decision support systems. The integration of electronic health records (EHRs), medical imaging data, and operational metrics has created unprecedented opportunities for data-driven healthcare improvements.

Recent studies have demonstrated that healthcare organizations utilizing advanced analytics report significant improvements in patient outcomes, operational efficiency, and cost management. The ability to identify patterns inpatient admissions, disease prevalence, and treatment effectiveness has become a cornerstone of modern healthcare management.

3.2 Role of Data Visualization in Healthcare

Data visualization plays a crucial role in making complex healthcare data accessible to clinical staff and administrators. Research has shown that well-designed dashboards can reduce decision-making time by up to 40% and improve the accuracy of clinical assessments by providing clear, intuitive representations of patient data.

The use of interactive dashboards in hospital settings has been particularly effective in monitoring key performance indicators, tracking patient flow, and identifying potential issues before they become critical problems. Studies indicate that hospitals using comprehensive dashboard systems report better patient satisfaction scores and improved operational metrics.

3.3 Tableau in Healthcare Settings

Tableau has emerged as a leading platform for healthcare data visualization due to its user-friendly interface, robust analytical capabilities, and ability to handle large datasets efficiently. Healthcare organizations worldwide have adopted Tableau for various applications, including patient monitoring, resource allocation, and quality improvement initiatives.

The platform's ability to create interactive, real-time dashboards has made it particularly valuable in clinical settings where timely access to accurate information is critical for patient care decisions.

4. Hospital Overview: Satguru Partap Singh Hospital

4.1 Institution Background

Satguru Partap Singh Hospital, located in Ludhiana, Punjab, stands as one of the region's premier healthcare institutions. The hospital has established itself as a leading provider of comprehensive medical services, serving a diverse patient population from across Punjab and neighbouring states.

The hospital's commitment to excellence in patient care is reflected in its state-of-the-art medical facilities, experienced healthcare professionals, and continuous adoption of modern healthcare technologies. The institution's focus on quality healthcare delivery makes it an ideal environment for exploring the applications of data analytics in clinical settings.

4.2 Hospital Infrastructure and Services

The hospital operates multiple specialized departments including cardiology, orthopaedics', paediatrics, general surgery, internal medicine, and emergency services. With a substantial bed capacity and advanced medical equipment, the hospital handles a significant volume of patient admissions daily, generating extensive data that requires systematic analysis and interpretation.

The hospital's information systems capture comprehensive patient data, including admission details, diagnostic information, treatment records, and outcome metrics. This rich dataset provides an excellent foundation for developing meaningful analytics and visualization projects.

4.3 Technology Infrastructure

The hospital has invested significantly in modern healthcare information systems, including electronic health records, hospital management systems, and clinical decision support tools. This technological infrastructure supports comprehensive data collection and provides the foundation for advanced analytics initiatives.

The hospital's commitment to leveraging technology for improved patient care creates an environment where data analytics projects can have immediate practical applications and measurable impact on healthcare delivery.

5. Objectives of the Project

5.1 Primary Objectives

The project was designed to achieve several key objectives that align with both academic learning goals and practical healthcare improvement initiatives:

Healthcare Domain Knowledge: Gain comprehensive understanding of healthcare data structures, common metrics, and key performance indicators used in hospital settings. Learn to interpret medical data accurately and identify meaningful patterns that can inform clinical decision-making.

5.2 Learning Outcomes

The project was structured to provide hands-on experience with real healthcare data while developing practical skills that are directly applicable to future career opportunities in healthcare analytics. The learning outcomes were designed to bridge the gap between theoretical knowledge and practical application.

Data Analysis Skills: Develop advanced analytical thinking skills through work with complex healthcare datasets. Learn to identify trends, patterns, and anomalies in patient data that can inform clinical and operational decisions.

Visualization Design: Master the principles of effective data visualization, including colour theory, layout design, and user experience considerations specific to healthcare applications.

Communication Skills: Enhance ability to communicate complex technical concepts to non-technical stakeholders, including healthcare professionals and hospital administrators.

5.3 Expected Impact

The project objectives were aligned with the hospital's strategic goals for improving patient care through data-driven insights. The expected impact included the development of dashboards that could be integrated into the hospital's regular operations to support ongoing decision-making processes.

The project was designed to demonstrate the value of data analytics in healthcare settings and provide a foundation for expanded analytics initiatives within the institution.

6. Methodology and Tools Used

6.1 Analytical Approach

The project project employed a systematic approach to healthcare data analytics, beginning with comprehensive data exploration and proceeding through structured visualization development. The methodology was designed to ensure accuracy, relevance, and practical applicability of all analytical outputs.

Iterative Development Process: Dashboard development followed an iterative approach, with regular feedback sessions from healthcare professionals to ensure that visualizations met practical needs and provided meaningful insights for clinical operations.

6.2 Technology Stack

Tableau Desktop: Primary visualization tool for creating interactive dashboards and analytical reports. Tableau's robust features enabled the development of sophisticated visualizations that could handle complex healthcare data relationships.

Data Sources: Hospital management systems, electronic health records, and administrative databases provided the foundation datasets for analysis. Data integration capabilities allowed for comprehensive analysis across multiple hospital systems.

Data Processing Tools: Various data cleaning and preparation tools were utilized to ensure data quality and consistency before visualization development.

6.3 Quality Assurance

Data Validation: Comprehensive validation processes were implemented to ensure accuracy of all data used in dashboard development.

Visualization Testing: All dashboards underwent extensive testing to ensure functionality, accuracy, and usability in real healthcare environments.

Performance Optimization: Dashboards were optimized for performance to ensure real-time responsiveness when accessing large healthcare datasets.

6.4 Key Performance Indicators (KPIs):

KPIs provide quick, high-level summaries of the most critical metrics related to hospital performance. In my dashboards, the following KPIs were used:

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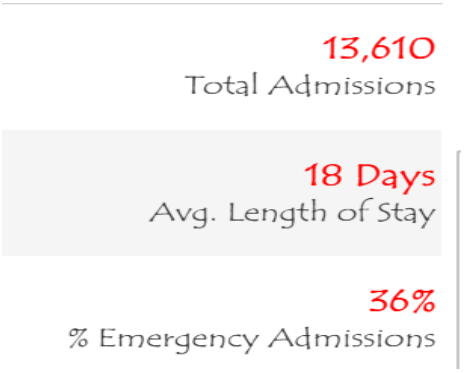
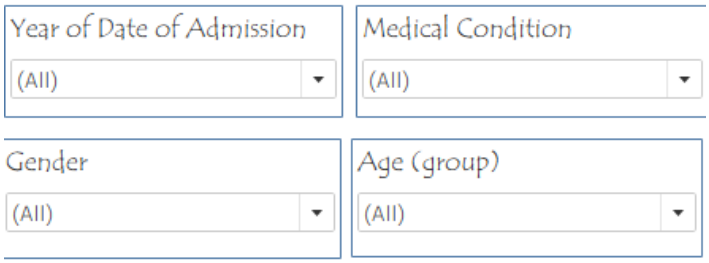
- **Total Number of Patients:** Displays the total number of unique patients in the dataset.
- **Average Length of Stay (LOS):** Shows the average number of days patients stayed in the hospital.
- **Percentage of Emergency Admissions:** Highlights what proportion of total admissions were emergencies.
- **Top Medical Condition:** Displays the most frequently occurring medical condition among all patients (based on year-wise filters).



6.5 Interactive Filters:

Filters were used to allow the user to dynamically explore data based on specific criteria. These filters help tailor the insights according to the user's requirements. Filters used include:

- **Year of Admission Filter:** Enables viewing trends year-wise.
- **Medical Condition Filter:** Lets users isolate specific conditions to examine patterns.
- **Age Group Filter:** Helps explore condition prevalence across different age categories.
- **Admission Type Filter:** Allows analysis of elective, emergency, routine, and urgent admissions.**Insurance Provider Filter:** Enables comparative



analysis of billing trends by insurance.

7. Data Collection and Processing

7.1 Data Sources and Types

The project project worked with a comprehensive healthcare dataset containing 14 key variables that collectively provided a holistic view of hospital operations and patient care metrics. The dataset structure included:

Patient Demographics: Name, Age, Gender, Blood Type - providing essential patient identification and demographic information for population health analysis.

Clinical Information: Medical Condition, Test Results, Medication - capturing core clinical data for disease pattern analysis and treatment effectiveness evaluation.

Admission Details: Date of Admission, Discharge Date, Admission Type, Room Number - enabling comprehensive patient flow and capacity utilization analysis.

Healthcare Provider Information: Doctor, Hospital - supporting provider-specific analysis and institutional benchmarking.

Financial Data: Insurance Provider, Billing Amount - enabling revenue analysis and insurance pattern evaluation.

This comprehensive dataset enabled multi-dimensional analysis across clinical, operational, and financial aspects of healthcare delivery, providing a foundation for evidence-based decision making and strategic planning.

Sr No	ID	Name	Age	Gende	Blood Type	Medical Condition	Date of Admission	Discharge Date	Doctor	Hospital	Insurance Provider	Billing Amount	Room Numbe	Admission Type	Length of Sta
1	1	Penny Phillips	35	Female	A+	Asthma	20/04/2021	26/04/2021	Aaron Aguilar	Loyola University Medical Centı	Aetna	3914	222	Emergency	6
2	2	Laura Rice	34	Female	A+	Infections	3/11/2021	12/11/2021	Aaron Aguilar	Loyola University Medical Centı	UnitedHealthcare	2944	384	Routine	9
3	3	Robin Kaufman	58	Female	O-	Cancer	28/12/2022	13/02/2023	Aaron Aguilar	Loyola University Medical Centı	Aetna	88013	459	Routine	47
4	4	Timothy Williams	53	Female	O+	Heart Disease	12/6/2019	26/06/2019	Aaron Aguilar	Loyola University Medical Centı	Medicare	64485	343	Emergency	14
5	5	Kristin Frazier	55	Male	O+	Cancer	24/11/2023	8/12/2023	Aaron Aguilar	Loyola University Medical Centı	Aetna	37479	247	Emergency	14
6	6	Travis Bennett	52	Female	O+	Obesity	7/1/2021	12/1/2021	Aaron Aguilar	Loyola University Medical Centı	Medicare	7468	377	Urgent	5
7	7	Lori Clark	59	Female	B-	Cancer	30/01/2023	20/03/2023	Aaron Aguilar	Loyola University Medical Centı	Medicare	41873	111	Elective	49
8	8	Kelsey Harris	41	Female	B-	Obesity	31/12/2019	8/1/2020	Aaron Aguilar	Loyola University Medical Centı	Cigna	8269	245	Urgent	8
9	9	Thomas Phelps	31	Female	O+	Infections	18/08/2019	23/08/2019	Aaron Aguilar	Loyola University Medical Centı	Blue Cross	1761	467	Emergency	5
10	10	Roy Barnett	60	Male	B+	Cancer	2/8/2020	16/09/2020	Aaron Aguilar	Loyola University Medical Centı	Blue Cross	77368	279	Elective	45
11	11	Brittany Gray	80	Female	A+	Heart Disease	3/11/2020	6/12/2020	Aaron Aguilar	Loyola University Medical Centı	Blue Cross	41340	196	Urgent	33
12	12	Robert Nichols	54	Female	A+	Cancer	18/05/2019	1/6/2019	Aaron Aguilar	Loyola University Medical Centı	Medicare	78624	353	Emergency	14
13	13	Michael Boyd	18	Female	O+	Asthma	8/12/2022	9/12/2022	Aaron Aguilar	Loyola University Medical Centı	Aetna	3434	235	Elective	1
14	14	Jill Silva	27	Male	B-	Flu	14/12/2023	18/12/2023	Aaron Aguilar	Loyola University Medical Centı	Aetna	1022	250	Emergency	4
15	15	Holly Mcintosh	58	Male	O+	Obesity	22/08/2021	28/08/2021	Aaron Aguilar	Loyola University Medical Centı	Cigna	13503	130	Routine	6
16	16	Joseph Garcia	15	Male	O+	Asthma	3/11/2019	8/11/2019	Aaron Aguilar	Loyola University Medical Centı	Aetna	6225	181	Emergency	5
17	17	Dr. Jennifer Thompson	61	Female	O+	Heart Disease	27/01/2022	20/02/2022	Aaron Aguilar	Loyola University Medical Centı	Aetna	27710	162	Elective	24
18	18	Kimberly Skinner Md	26	Female	A+	Asthma	28/04/2023	4/5/2023	Aaron Aguilar	Loyola University Medical Centı	UnitedHealthcare	7312	296	Urgent	6
19	19	Adam Orr	46	Female	O+	Obesity	16/03/2020	22/03/2020	Aaron Aguilar	Loyola University Medical Centı	Aetna	12610	275	Elective	6
20	20	Devon Zhang	10	Female	O+	Flu	14/01/2024	18/01/2024	Aaron Aguilar	Loyola University Medical Centı	Blue Cross	4496	168	Routine	4
21	21	Bobby Todd	27	Male	AB-	Flu	24/05/2023	25/05/2023	Aaron Aguilar	Loyola University Medical Centı	Blue Cross	2782	175	Emergency	1
22	22	Jessica Elliott Dvm	51	Female	B-	Diabetes	12/4/2023	24/04/2023	Aaron Aguilar	Loyola University Medical Centı	UnitedHealthcare	17797	457	Elective	12
23	23	Tony Colon	11	Male	O+	Flu	4/12/2023	6/12/2023	Aaron Aguilar	Loyola University Medical Centı	Cigna	1796	479	Emergency	2
24	24	Christine Thompson	28	Female	O+	Asthma	6/11/2020	12/11/2020	Aaron Aguilar	Loyola University Medical Centı	Medicare	2574	206	Emergency	6
25	25	Bruce York	64	Female	O+	Alzheimerâ€™s	28/09/2020	1/11/2020	Aaron Aguilar	Loyola University Medical Centı	Aetna	16408	310	Routine	34
26	26	Samantha Patterson	45	Female	A+	Obesity	11/9/2022	18/09/2022	Aaron Aguilar	Loyola University Medical Centı	Medicare	5996	303	Elective	7
27	27	Stephen Malone	62	Female	B+	Heart Disease	14/04/2023	5/5/2023	Aaron Aguilar	Loyola University Medical Centı	UnitedHealthcare	35132	464	Emergency	21
28	28	Katrina Cook	68	Male	AB+	Alzheimerâ€™s	14/02/2023	21/04/2023	Aaron Aguilar	Loyola University Medical Centı	Medicare	21269	440	Emergency	66
29	29	Tammy Mills	43	Male	AB+	Diabetes	29/03/2022	2/4/2022	Aaron Aguilar	Loyola University Medical Centı	Cigna	11131	448	Elective	4
30	30	Cynthia Frye	27	Female	A+	Flu	29/06/2022	2/7/2022	Aaron Aguilar	Loyola University Medical Centı	Medicare	2422	305	Emergency	3
31	31	Pamela Jackson	64	Female	AB-	Cancer	2/12/2023	18/12/2023	Aaron Aguilar	Loyola University Medical Centı	Cigna	82938	385	Routine	16
32	32	Steven Barker	52	Female	A+	Obesity	14/04/2022	17/04/2022	Aaron Aguilar	Loyola University Medical Centı	Cigna	6883	438	Routine	3
33	33	Aaron Lopez	73	Female	B-	Heart Disease	17/01/2022	30/01/2022	Aaron Aguilar	Loyola University Medical Centı	UnitedHealthcare	57943	139	Urgent	13
34	34	Zachary Dixon	80	Female	B+	Cancer	5/2/2021	3/3/2021	Aaron Aguilar	Loyola University Medical Centı	Aetna	36644	334	Routine	26
35	35	Charles Parker	71	Male	A+	Heart Disease	11/10/2020	29/10/2020	Aaron Aguilar	Loyola University Medical Centı	Medicare	49589	391	Emergency	18
36	36	Heather Lee	37	Female	A+	Flu	2/9/2021	6/9/2021	Aaron Aguilar	Loyola University Medical Centı	Aetna	2880	197	Urgent	4
37	37	Mark Norton	8	Male	O+	Flu	8/5/2022	9/5/2022	Aaron Aguilar	Loyola University Medical Centı	Aetna	4976	327	Emergency	1
38	38	Kathleen Russell	65	Male	O+	Alzheimerâ€™s	2/3/2024	4/4/2024	Aaron Aguilar	Loyola University Medical Centı	Blue Cross	24751	167	Elective	33

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	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Sr N	I	Name	Ag	Gend	Blood Ty	Medical Conditio	Date of Admissio	Discharge Da	Doctor	Hospital	Insurance Provid	Billing Amou	Room Numb	Admission Ty
13	812	812	Richard Stein Jr.	85	Male	O+	Alzheimerâ€™s	11/7/2022	8/8/2022	Angela Mitchell	Loyola University Medical Center	Blue Cross	17995	393	Elective
26	825	825	Jose Thomas	56	Male	B-	Cancer	9/8/2020	23/08/2020	Angela Mitchell	Loyola University Medical Center	Blue Cross	51315	270	Elective
37	836	836	Emily Reese	64	Male	AB+	Cancer	11/1/2021	18/02/2021	Angela Mitchell	Loyola University Medical Center	Cigna	65413	261	Elective
45	844	844	Jose Young	65	Male	B-	Cancer	25/02/2020	17/04/2020	Angela Mitchell	Loyola University Medical Center	UnitedHealthcare	62651	248	Elective
51	850	850	Marissa Vaughan	52	Male	A-	Cancer	9/2/2020	25/03/2020	Angela Mitchell	Loyola University Medical Center	Cigna	34152	136	Elective
64	863	863	Jessica Adams	66	Male	O-	Cancer	21/06/2022	19/08/2022	Angela Mitchell	Loyola University Medical Center	Cigna	60915	494	Urgent
83	882	882	Kenneth Powell	75	Male	A+	Cancer	19/08/2020	5/10/2020	Angela Mitchell	Loyola University Medical Center	Aetna	48231	402	Emergency
86	885	885	Michael Walker	69	Male	AB-	Alzheimerâ€™s	5/1/2022	2/2/2022	Angela Mitchell	Loyola University Medical Center	Medicare	23409	290	Routine
90	889	889	John Thompson	74	Male	O-	Alzheimerâ€™s	29/05/2019	21/06/2019	Angela Mitchell	Loyola University Medical Center	Aetna	38777	396	Urgent
91	890	890	Brian Garza	59	Male	B-	Cancer	7/1/2021	7/3/2021	Angela Mitchell	Loyola University Medical Center	Medicare	43495	215	Urgent
01	900	900	Brenda Stevenson	52	Male	B+	Cancer	12/8/2022	10/10/2022	Angela Mitchell	Loyola University Medical Center	Aetna	93210	173	Elective
17	916	916	Brandon Martinez	52	Male	A+	Cancer	28/06/2021	5/8/2021	Angela Mitchell	Loyola University Medical Center	Aetna	86166	121	Elective
28	927	927	Angela Brown	55	Male	AB+	Cancer	16/03/2021	1/5/2021	Angela Mitchell	Loyola University Medical Center	Cigna	53204	370	Elective
29	928	928	Morgan Harmon	51	Male	A+	Cancer	20/12/2020	2/2/2021	Angela Mitchell	Loyola University Medical Center	Cigna	51388	444	Routine
30	929	929	Tiffany Hayes	79	Male	A+	Cancer	16/06/2020	11/8/2020	Angela Mitchell	Loyola University Medical Center	Blue Cross	82602	155	Elective
37	936	936	Caleb Cox	65	Male	B+	Alzheimerâ€™s	13/05/2020	7/7/2020	Angela Mitchell	Loyola University Medical Center	Cigna	17348	167	Routine
52	951	951	Stacey Anderson	55	Male	AB-	Cancer	10/4/2020	25/05/2020	Angela Mitchell	Loyola University Medical Center	Medicare	82116	282	Urgent
57	956	956	Dr. Vanessa Taylor Md	72	Male	O+	Cancer	10/3/2022	9/4/2022	Angela Mitchell	Loyola University Medical Center	Medicare	60775	392	Elective
66	965	965	Sharon Lee	62	Male	O+	Cancer	22/08/2020	14/10/2020	Angela Mitchell	Loyola University Medical Center	Medicare	56366	468	Routine
70	969	969	Jason Morton	69	Male	B-	Cancer	5/1/2021	14/02/2021	Angela Mitchell	Loyola University Medical Center	Cigna	95966	496	Urgent
89	988	988	Katie Henderson	83	Male	B+	Alzheimerâ€™s	9/7/2023	29/09/2023	Angela Mitchell	Loyola University Medical Center	Aetna	16593	232	Emergency
96	995	995	Karla Lewis	69	Male	O-	Alzheimerâ€™s	31/10/2020	20/11/2020	Angela Mitchell	Loyola University Medical Center	Cigna	45880	234	Routine
09	1008	8	Kelsey Harris	61	Male	A+	Cancer	10/4/2024	11/5/2024	Angela Mitchell	Loyola University Medical Center	Medicare	32074	362	Routine
10	1009	9	Thomas Phelps	70	Male	A+	Cancer	17/02/2023	12/3/2023	Angela Mitchell	Loyola University Medical Center	Blue Cross	70026	489	Elective
15	1014	14	Jill Silva	82	Male	O+	Alzheimerâ€™s	31/03/2023	24/06/2023	Angela Mitchell	Loyola University Medical Center	Blue Cross	38766	257	Urgent
26	1025	25	Bruce York	85	Male	A-	Alzheimerâ€™s	6/12/2022	6/2/2023	Angela Mitchell	Loyola University Medical Center	UnitedHealthcare	31497	348	Elective
37	1036	36	Heather Lee	90	Male	O+	Alzheimerâ€™s	9/7/2020	27/09/2020	Angela Mitchell	Loyola University Medical Center	Cigna	38896	195	Emergency

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8. Dashboard Development

8.1 Admissions Statistics Dashboard

The Admissions Statistics Dashboard was developed to provide comprehensive insights into patient admission patterns, helping hospital administrators understand utilization trends and optimize resource allocation.

Key Metrics and Visualizations:

Admission Type Analysis by Medical Condition: The primary visualization displays the distribution of different medical conditions (Asthma, Diabetes, Heart Disease, Obesity) across various admission types (Elective, Emergency, Routine, Urgent). This analysis revealed that:

- Elective admissions are highest for routine procedures like obesity management (637 cases)
- Emergency admissions show significant volumes for acute conditions like asthma (622 cases)
- Urgent care admissions demonstrate consistent patterns across cardiovascular and metabolic conditions
- The color-coded visualization enables quick identification of admission patterns and resource planning needs

Financial Analysis by Insurance Provider: The billing amount analysis across different insurance providers (Aetna, Blue Cross, Cigna, Medicaid, United Healthcare) shows:

- Total billing amounts ranging from approximately 57M to 62M across major providers
- Blue Cross and Cigna represent the highest billing volumes (approximately 61-62M each)
- Medicaid and United Healthcare show lower but significant billing amounts (58-59M)
- This analysis supports financial planning and insurance contract negotiations

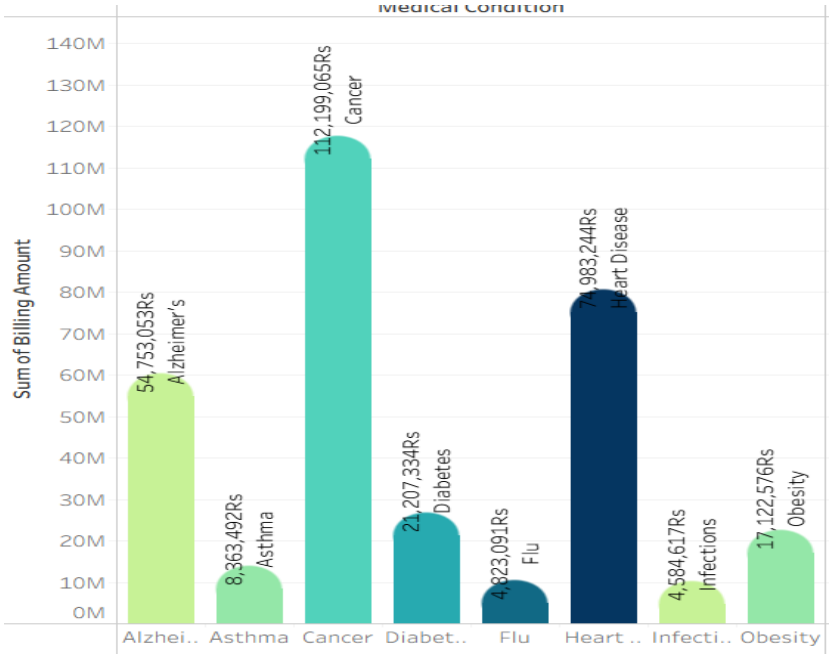
Temporal Billing Trends: The date-wise admission billing analysis reveals:

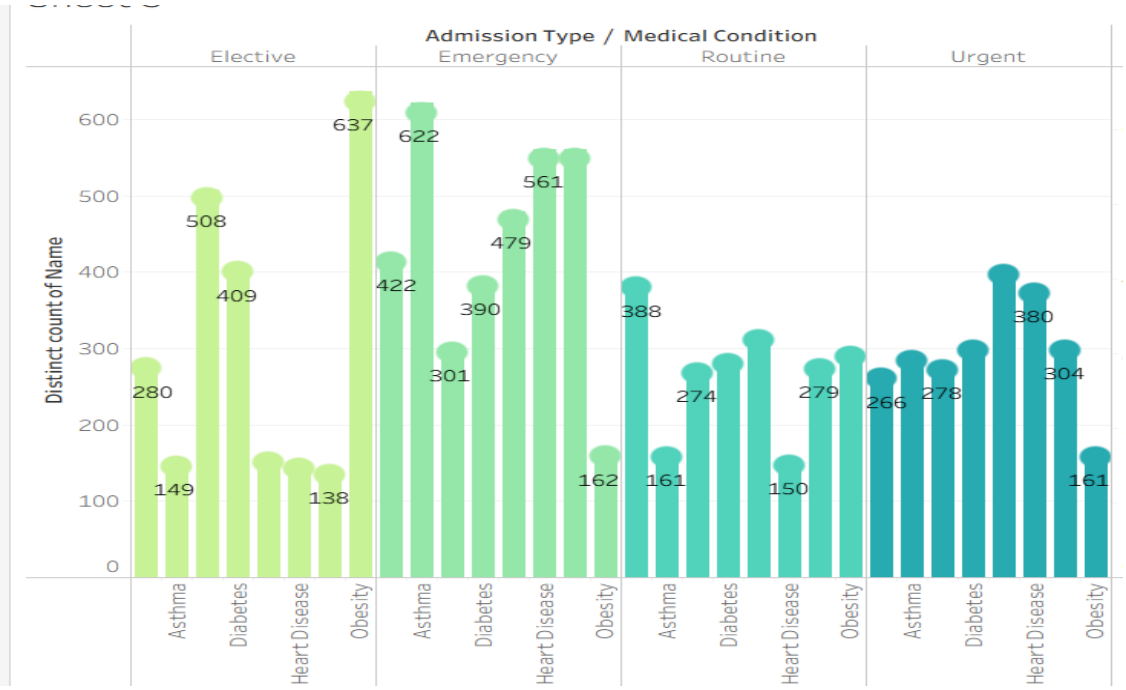
- Peak billing periods in 2020-2021 (approximately 60M)
- Steady decline in 2022-2023 billing amounts
-

- Significant drop in 2024 (21.95M), indicating seasonal patterns or operational changes
- These trends inform capacity planning and revenue forecasting

Design Principles and User Experience:

The dashboard utilizes a clean, professional design with distinct colour coding for different categories. Interactive elements allow users to filter by specific conditions or time periods. The visualizations are optimized for quick decision-making, with key metrics prominently displayed and supporting details accessible through drill-down functionality.





8.2 Disease Statistics Dashboard

The Disease Statistics Dashboard provided comprehensive analysis of disease patterns and diagnostic trends within the hospital population, supporting clinical decision-making and public health initiatives.

Clinical Analysis Components:

Medical Condition Distribution by Age Groups: The comprehensive age-based analysis (0-15, 16-35, 36-55, 56-65, 66+ years) across medical conditions reveals critical epidemiological patterns:

- Alzheimer's Disease: Predominantly affects elderly populations (66+ age group with 729 cases), with minimal occurrence in younger demographics
- Asthma: Shows highest prevalence in the 16-35 age group (671 cases), indicating the need for targeted respiratory care programs
- Cancer: Demonstrates age-progressive increase, with highest cases in 36-55 (571 cases) and 56-65 (668 cases) age groups
- Diabetes: Peaks in the 56-65 age group (433 cases) and remains significant in elderly populations
- Heart Disease: Shows consistent high prevalence across middle-aged and elderly populations (404-546 cases)
- Infections: Highest in paediatric (0-15: 284 cases) and young adult (16-35: 521 cases) populations
- Obesity: Distributed across all age groups with notable peaks in 36-55 (293 cases) and 66+ (252 cases) demographics

Disease Pattern Insights: The analysis reveals important epidemiological trends:

- Age-related chronic conditions (Alzheimer's, Heart Disease) show expected demographic distributions
- Lifestyle-related conditions (Diabetes, Obesity) demonstrate multi-generational impact
- Infectious diseases predominantly affect younger populations, requiring targeted prevention strategies
- Cancer shows mid-life prevalence patterns, supporting screening program development

Resource Allocation Implications: The age-stratified disease analysis provides crucial information for:

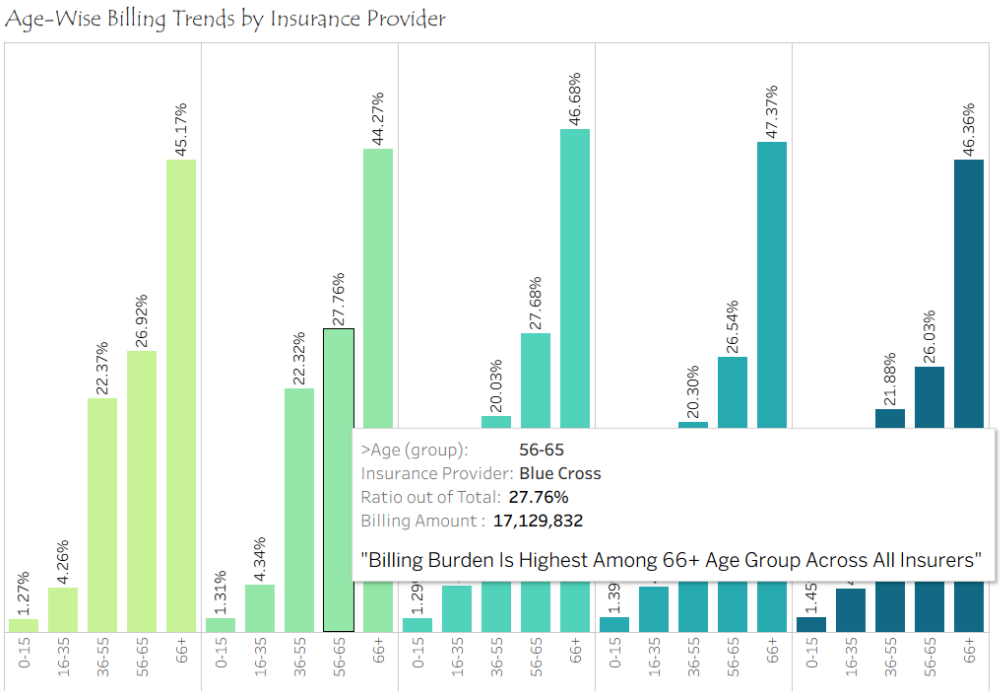
- Specialist staffing requirements across different age-focused departments

- Equipment and facility planning for age-specific medical needs
- Preventive care program development targeting high-risk demographics
- Pharmaceutical inventory management based on age-related medication needs

Public Health Integration:

The dashboard incorporates broader public health perspectives, comparing hospital disease patterns with regional and national health statistics. This integration provides context for local health trends and supports the hospital's role in community health improvement.

The visualization design emphasizes the clinical relevance of data, using medical terminology appropriately while maintaining accessibility for various healthcare professionals. Interactive features allowed clinicians to explore specific disease categories and investigate patterns relevant to their specialties.



9. Key Findings and Insights

9.1 Admission Pattern Analysis

The analysis of admission patterns revealed several significant insights that have direct implications for hospital operations and resource management:

Temporal Trends: Clear patterns emerged in admission timing, with distinct peaks during certain days of the week and seasonal variations. Monday admissions showed consistently higher volumes, likely reflecting weekend healthcare needs and scheduled procedures. Monthly analysis revealed seasonal patterns with increased admissions during monsoon months due to waterborne diseases and respiratory conditions.

Demographic Insights: The patient population analysis revealed important demographic patterns that inform service planning. The age distribution showed a significant representation of elderly patients, highlighting the need for geriatric care specialization. Gender-based analysis revealed specific healthcare needs that could be addressed through targeted service development.

Department Utilization: Analysis of department-wise admissions identified high-volume areas requiring additional resources and departments with capacity for expansion. Emergency department utilization patterns provided insights into community health needs and optimal staffing requirements.

9.2 Disease Pattern Analysis

The disease statistics analysis provided valuable insights into the health profile of the hospital's patient population:

Prevalence Patterns: The analysis identified the most common diseases and conditions treated at the hospital, providing a foundation for resource allocation and specialist recruitment decisions. Cardiovascular diseases, diabetes, and respiratory conditions emerged as significant health challenges requiring focused attention.

Comorbidity Analysis: The examination of multiple conditions in single patients revealed important patterns of disease association, informing holistic treatment approaches and supporting coordinated care initiatives.

Treatment Effectiveness: Analysis of treatment outcomes across different disease categories provided insights into clinical effectiveness and identified opportunities for protocol improvement and best practice implementation.

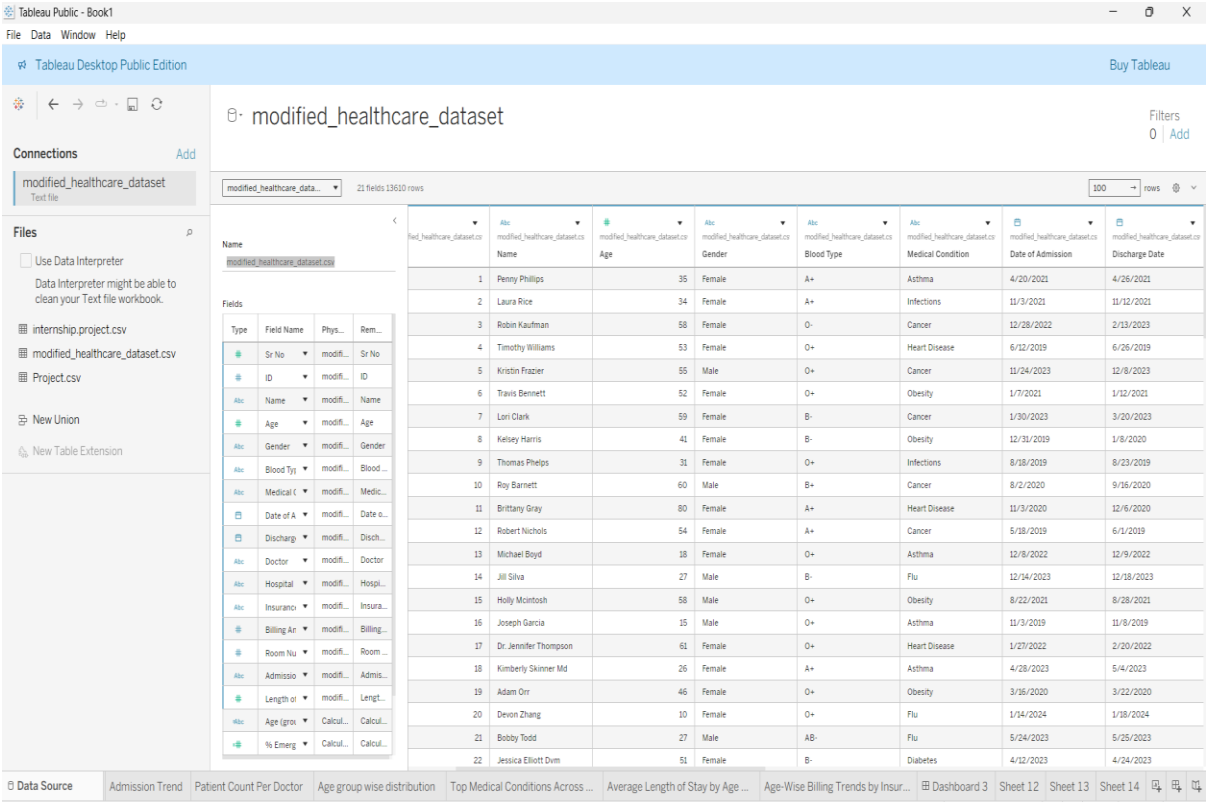
9.3 Operational Insights

The combined analysis of admission and disease data provided important operational insights:

Resource Optimization: The analysis identified opportunities for more efficient resource allocation, including bed management optimization and staff scheduling improvements based on predictable patient flow patterns.

Capacity Planning: Understanding of peak utilization periods and seasonal variations supported strategic capacity planning and infrastructure development decisions.

Quality Improvement: The identification of variation in treatment outcomes across different conditions provided a foundation for quality improvement initiatives and clinical protocol enhancement.



10. Data Visualizations and Dashboard Screenshots

The following section presents the visual components developed for this project, including key data visualizations and dashboard screenshots that support the analytical findings discussed throughout this report.

10.1 Admission Statistics Dashboard

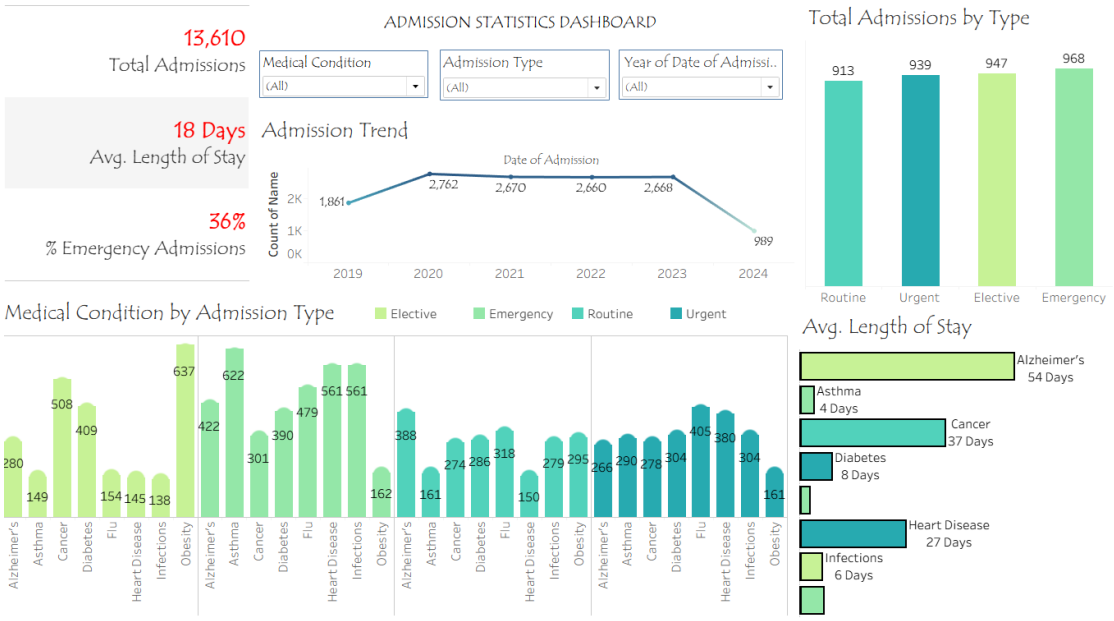


Figure 10.1: Dashboard showing admission trends by day of week, month, and seasonal patterns

10.2 Insurance Provider Distribution

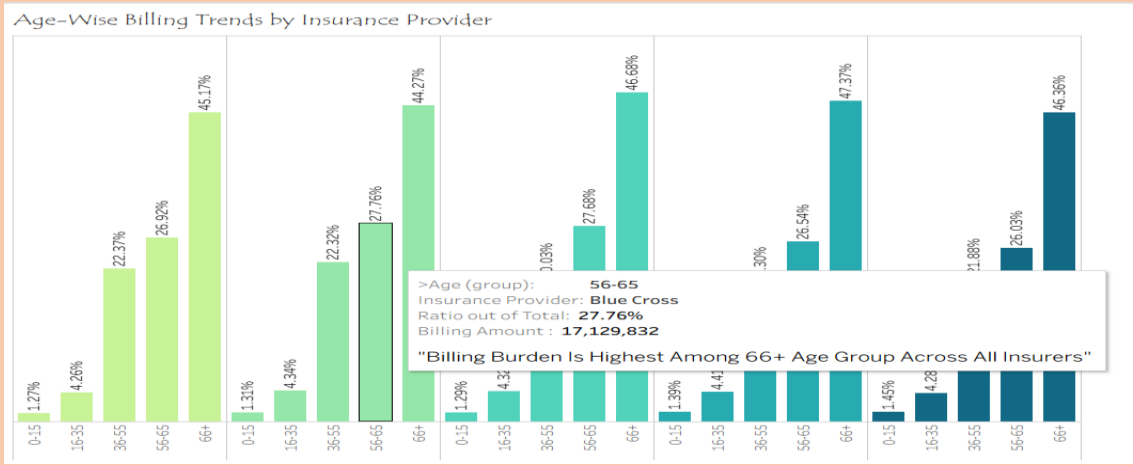


Figure 10.2: Visualization of insurance coverage patterns across different demographic segment

10.3 Disease Statistics Dashboard

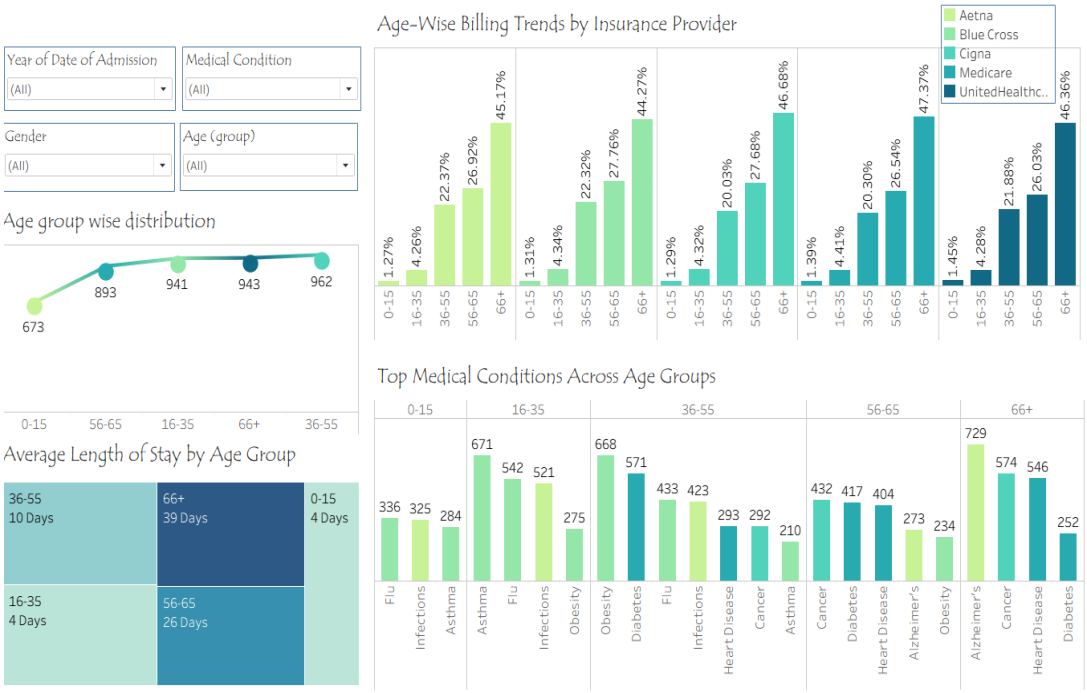


Figure 10.3: Comprehensive visualization of disease patterns across age groups and conditions

10.4 Stay Utilization Heatmap

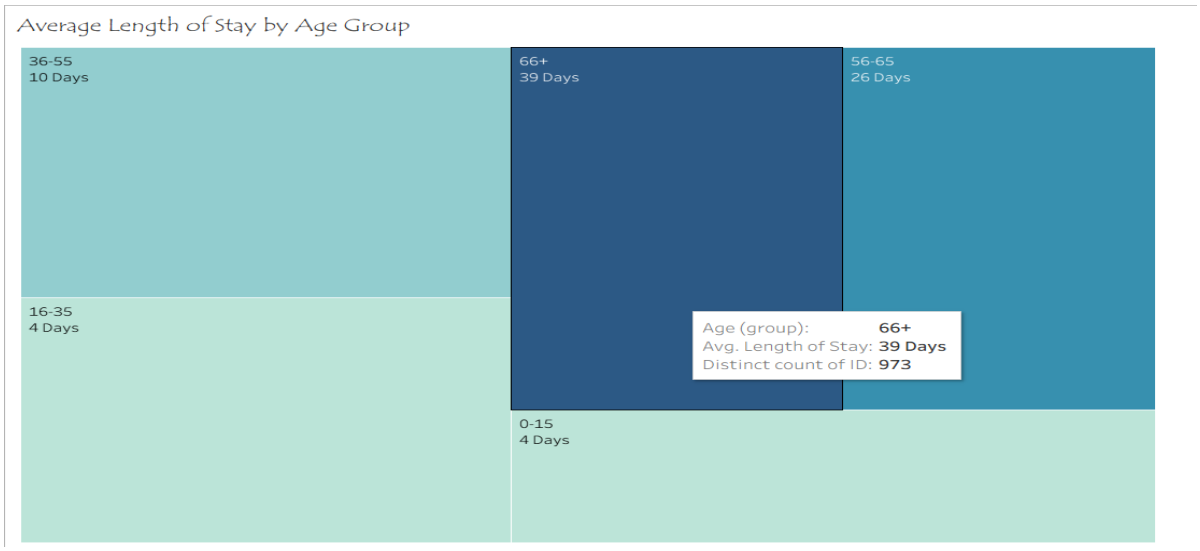


Figure 10.4: Heatmap showing intensity of Length of stay by time period and patient count

11. Challenges Faced and Solutions

During the course of this project report on hospital data analytics using Tableau, several technical and conceptual challenges were encountered. Below is a summary of the key obstacles and the solutions adopted to overcome them:

1. Understanding the Healthcare Dataset

Challenge:
Initially, it was difficult to understand the relevance and relationship between fields such as Admission Type, Length of Stay, Insurance, and Medical Condition.

Solution:
I held discussions with mentors and studied the structure of hospital operations to understand the real-world meaning of each field. This helped in deciding how to use dimensions and measures effectively.

2. Handling Date Formats and Calculated Fields

Challenge:
Date fields like "Date of Admission" and "Discharge Date" were not in a consistent format. Calculating Length of Stay was also confusing in Tableau initially.

Solution:
I reformatted date fields and created calculated fields in Tableau to determine metrics such as Length of Stay. For example:
DATEDIFF ('day', [Date of Admission], [Discharge Date])

3. Visual Overlap and Clutter

Challenge:
When visualizing multiple metrics (e.g., medical condition by age group and admission type), bar charts became overcrowded and hard to read.

Solution:
I used combinations of visuals like side-by-side bars, dual axes, and highlight tables. I also applied colour coding and interactive filters to reduce clutter and improve readability.

12. The Future of Healthcare Analytics

This project represents the initial step toward building a comprehensive hospital analytics platform. While the current focus is on admissions, doctor performance, and room utilization, there is significant scope to expand the system into a complete decision-support tool for healthcare management.

In the future, the dashboard can be enhanced by integrating real-time patient monitoring data, allowing administrators to track room availability and patient status live. Predictive analytics can also be implemented to forecast readmission risks, optimize resource allocation, and estimate average length of stay for different conditions.

Further, this dashboard can incorporate financial data such as insurance claims, billing, and reimbursement tracking, allowing for deeper insights into the hospital's financial health. Performance metrics for doctors, departments, and procedures can also be added to evaluate outcomes and service quality.

With advancements in AI and machine learning, this project could evolve into an intelligent hospital analytics system capable of generating recommendations, alerts, and automated insights. As hospitals continue to digitize records, such tools will play a vital role in enhancing patient care, improving workflows, and supporting strategic planning at every level of healthcare management.

13. Conclusion

The project at **Chandigarh University** offered a transformative experience in applying data analytics within the healthcare domain. Through hands-on exposure to real-world data, I learned how to clean, analyse, and visualize insights using Tableau. The project covered a wide range of hospital operational metrics such as admission trends, doctor workload, room utilization, average length of stay, billing distribution, and age-wise disease patterns.

By building interactive dashboards, I translated raw data into meaningful visualizations that can assist hospitals in identifying bottlenecks, optimizing resources, and improving patient outcomes. The most valuable learning was understanding how data-driven decisions can streamline hospital operations and enhance care delivery. This project not only strengthened my technical skills but also instilled a deep appreciation for how analytics can contribute to public health systems. Going forward, the project can be expanded further to include predictive analytics and real-time data monitoring, making it an essential tool for healthcare institutions.

14. Evaluation Grid

S.no	Parameters	Marks Obtained	Maximum Marks
1.	Project Title		2 Mark
2.	Design & Implementation		5 Mark
3.	GitHub Link		1 Mark
4.	LinkedIn Blog Link		1 Mark
5.	Portfolio Link		1 Mark
	TOTAL		10 Mark
	AVG		6 Mark